

CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY
OPERATING SYSTEMS (CT104H)
LAB #3

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- **Submission:** Students submit 1 file named *StudentName_ID_CT104H_Lab03.pdf* to the Google classroom (where *StudentName* is the student's name, and *ID* is the student's ID).

- **Instructions on how to present in the report file**

☐ For each question, students MUST provide the commands/scripts AND screenshots of the commands used and/or the content of files/scripts, CLEARLY. Note: *the screenshot needs to include the name of the Ubuntu Virtual machine.*

☐ Student creates an Ubuntu Virtual machine named *UbuntuID* (*ID* is the student's ID). For example, we have a virtual machine named **Ubuntu20043**, such that the student *ID* is **20043**.

Question 0: Navigate to your home directory

Answer: \$cd



Study from the files: Linux Shell Scripting Tutorial, and Shell Script (Chapter 2, 3), Advanced Bash-Scripting Guide (Chapter 13)

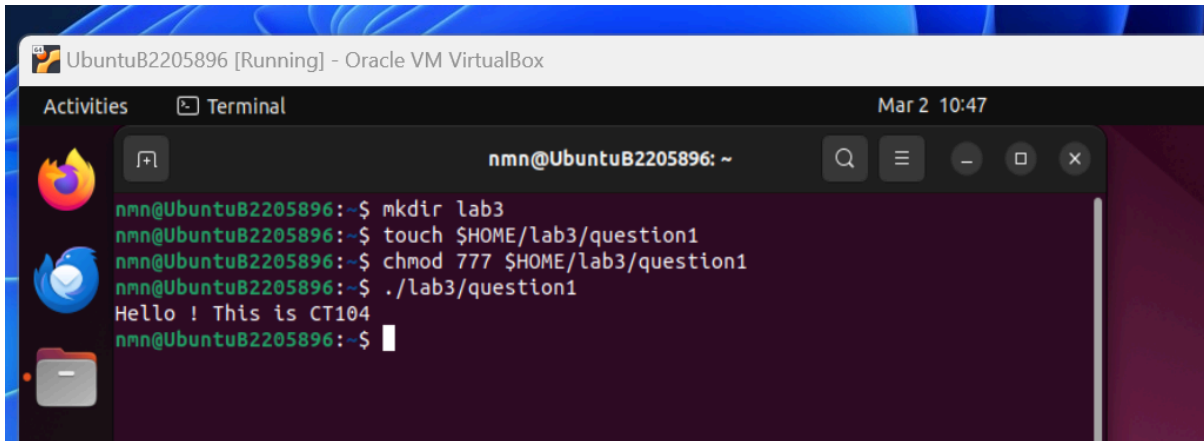
Question 1: Write a shell script that prints the message “Hello ! This is CT104” to the screen.

Answer:

```
#!/bin/bash
```

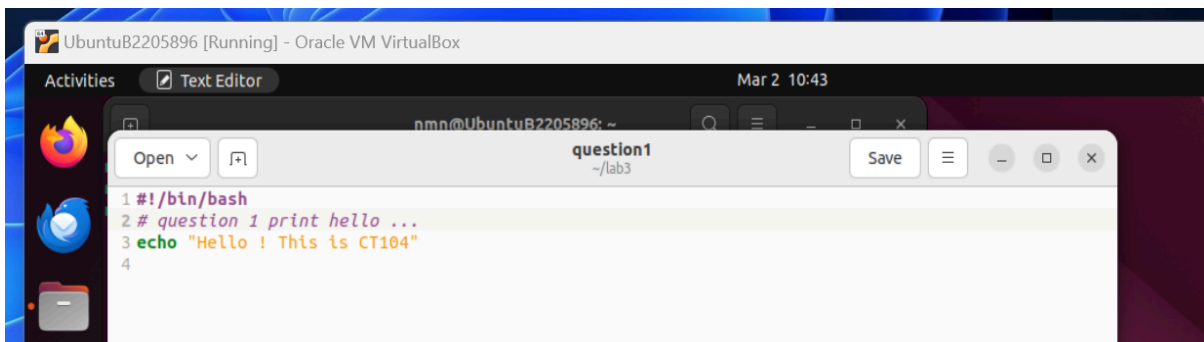
```
# question 1 print hello
```

```
echo “Hello! This is CT104”
```



The screenshot shows a terminal window titled "UbuntuB2205896 [Running] - Oracle VM VirtualBox". The terminal output is as follows:

```
nmn@UbuntuB2205896:~$ mkdir lab3
nmn@UbuntuB2205896:~$ touch $HOME/lab3/question1
nmn@UbuntuB2205896:~$ chmod 777 $HOME/lab3/question1
nmn@UbuntuB2205896:~$ ./lab3/question1
Hello ! This is CT104
nmn@UbuntuB2205896:~$
```



The screenshot shows a text editor window titled "question1 ~/lab3". The content of the file is as follows:

```
1 #!/bin/bash
2 # question 1 print hello ...
3 echo "Hello ! This is CT104"
4
```

Question 2: Modify the shell script from Question 1 to include a variable. The variable will hold the content of the message.

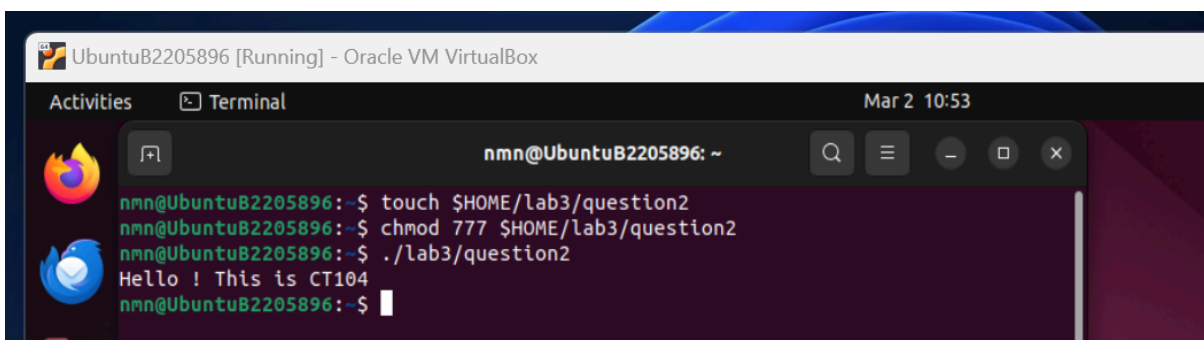
Answer:

```
#!/bin/bash
```

```
# question 1 print hello ... with variable
```

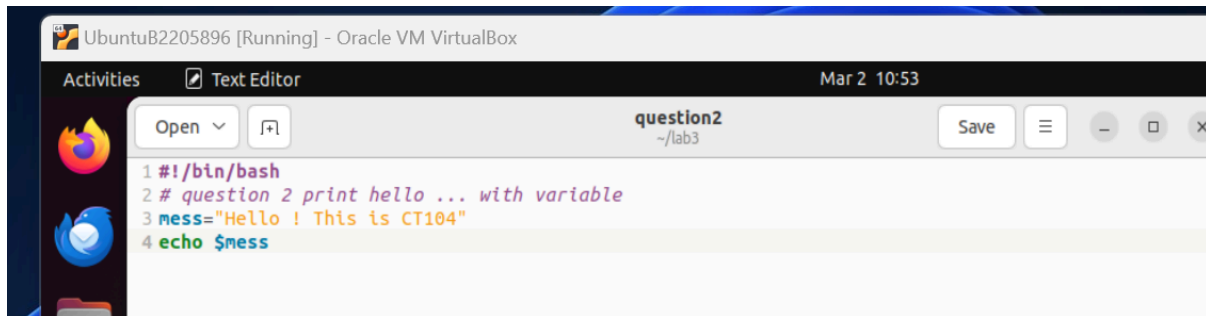
```
mess=“Hello! This is CT104”
```

```
echo $mess
```



The screenshot shows a terminal window titled "UbuntuB2205896 [Running] - Oracle VM VirtualBox". The terminal output is as follows:

```
nmn@UbuntuB2205896:~$ touch $HOME/lab3/question2
nmn@UbuntuB2205896:~$ chmod 777 $HOME/lab3/question2
nmn@UbuntuB2205896:~$ ./lab3/question2
Hello ! This is CT104
nmn@UbuntuB2205896:~$
```



The screenshot shows a text editor window titled 'question2' with the following content:

```
1 #!/bin/bash
2 # question 2 print hello ... with variable
3 mess="Hello ! This is CT104"
4 echo $mess
```

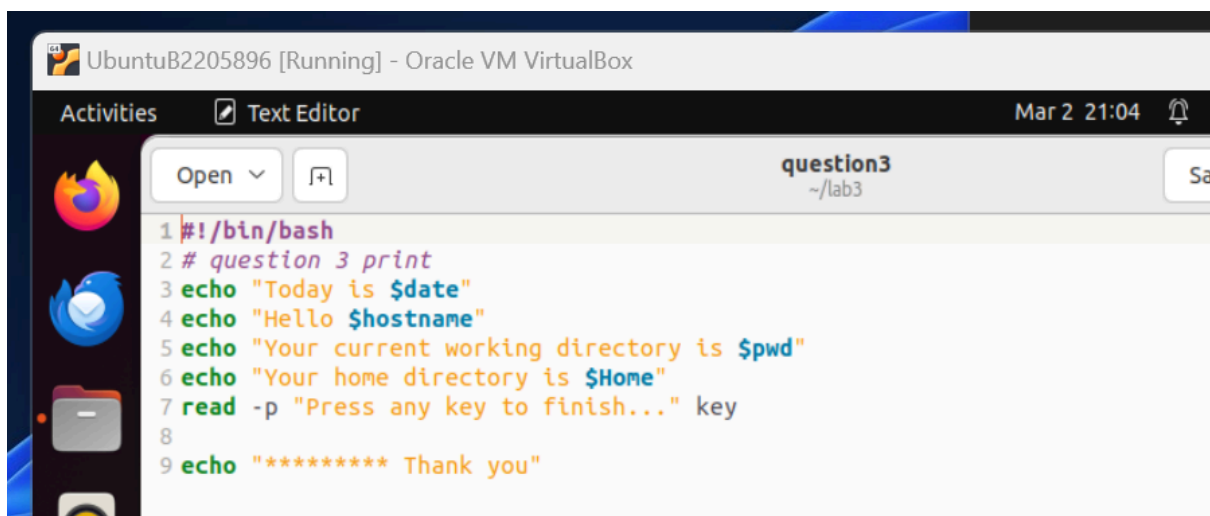
Question 3: Write a shell script that prints the following information to the screen

```
Today is <current date>
Hello <username>
Your current working directory is <current directory>
Your home directory is <home directory>
Press any key to finish <waiting to receive a key from user>

***** Thank you
```

Answer:

```
#!/bin/bash
# question 3 print
echo "Today is $date"
echo "Hello $hostname"
echo "Your current working directory is $pwd"
echo "Your home directory is $Home"
read -p "Press any key to finish..." key
echo "***** Thank you"
```



The screenshot shows a text editor window titled 'question3' with the following content:

```
1 #!/bin/bash
2 # question 3 print
3 echo "Today is $date"
4 echo "Hello $hostname"
5 echo "Your current working directory is $pwd"
6 echo "Your home directory is $Home"
7 read -p "Press any key to finish..." key
8
9 echo "***** Thank you"
```

```

nmn@UbuntuB2205896 [Running] - Oracle VM VirtualBox
Activities Terminal Mar 2 11:22
nmn@UbuntuB2205896: ~
nmn@UbuntuB2205896:~$ touch $HOME/lab3/question3
nmn@UbuntuB2205896:~$ chmod 777 $HOME/lab3/question3
nmn@UbuntuB2205896:~$ ./lab3/question3
Today is
Hello
Your current working directory is
Your home directory is
Press any key to finish...t
***** Thank you
nmn@UbuntuB2205896:~$
  
```

Question 4: Write a shell script that receives 2 numbers from a user, and calculates the following values

```

Please input the first number: x = x
Please input the second number: y = y
= <x-y>
) = <x+y>
) = <x*y>
= <x/y>
) = <x%y>
2
x!
  
```

Answer:

```

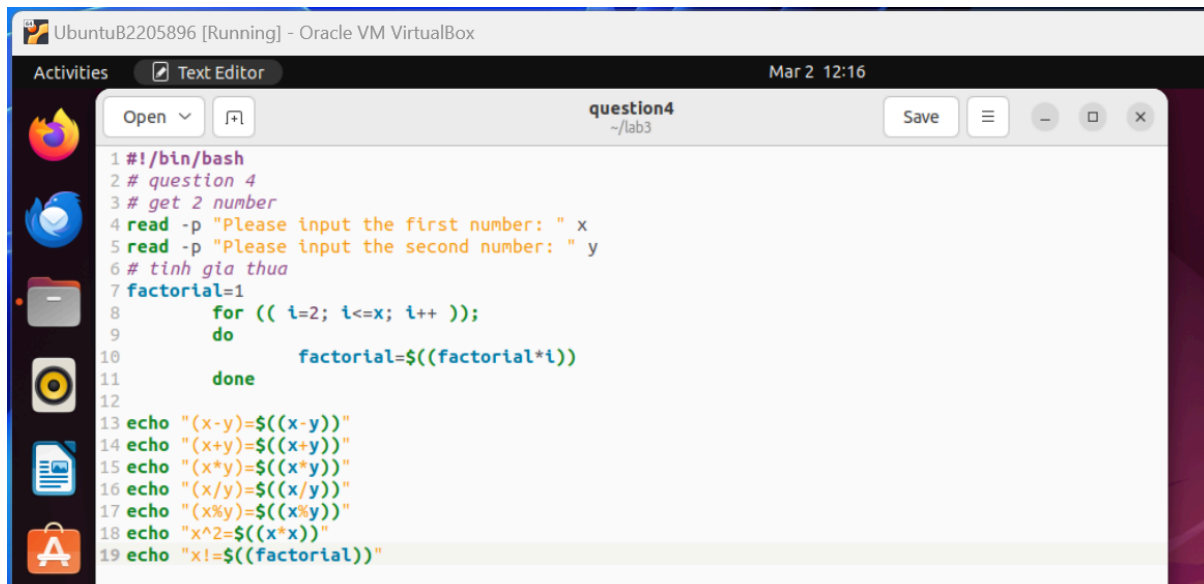
#!/bin/bash
# question 4
# get 2 number
read -p "Please input the first number: "x
read -p "Please input the second number: y
# tinh gia thua
factorial=1

for (i=2; i<=x; i++ );
do
factorial=$((factorial*i))
done
echo "(x-y)=$((x-y))"
echo "(x+y)=$((x+y))"
  
```

```

echo "(x*y)=$((x*y))"
echo "(x/y)=$((x/y))"
echo "(xy)=$((x%y))"
echo "x^2=$((x*x))"
echo "x!=$((factorial))"

```

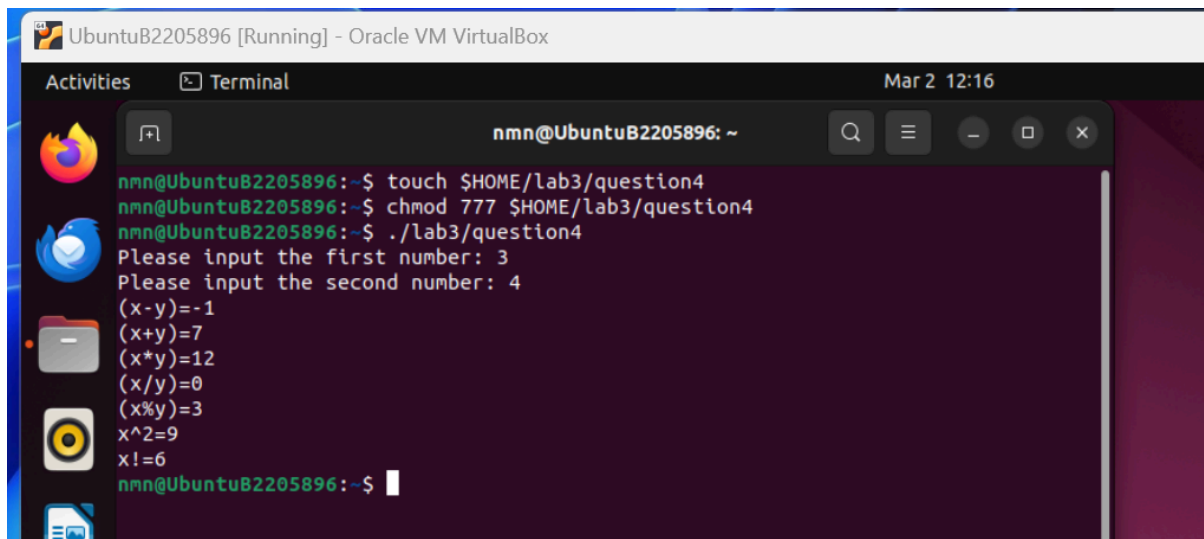


The screenshot shows a text editor window titled 'question4' in the Oracle VM VirtualBox environment. The script content is as follows:

```

1 #!/bin/bash
2 # question 4
3 # get 2 number
4 read -p "Please input the first number: " x
5 read -p "Please input the second number: " y
6 # tinh gia thua
7 factorial=1
8   for (( i=2; i<=x; i++ ));
9     do
10       factorial=$((factorial*i))
11     done
12
13 echo "(x-y)=$((x-y))"
14 echo "(x+y)=$((x+y))"
15 echo "(x*y)=$((x*y))"
16 echo "(x/y)=$((x/y))"
17 echo "(x%y)=$((x%y))"
18 echo "x^2=$((x*x))"
19 echo "x!=$((factorial))"

```



The screenshot shows a terminal window in the Oracle VM VirtualBox environment. The user has created the script, set permissions, and executed it. The output is as follows:

```

nmn@UbuntuB2205896:~$ touch $HOME/lab3/question4
nmn@UbuntuB2205896:~$ chmod 777 $HOME/lab3/question4
nmn@UbuntuB2205896:~$ ./lab3/question4
Please input the first number: 3
Please input the second number: 4
(x-y)=-1
(x+y)=7
(x*y)=12
(x/y)=0
(x%y)=3
x^2=9
x!=6
nmn@UbuntuB2205896:~$

```

Question 5: Write a shell script that allows a user to input a name of a directory, and creates this directory for the user. If the directory is created, please print the message “The *name_of_directory* is created successfully” to the screen; otherwise, print the message “Can not create the *name_of_directory*” to the screen.

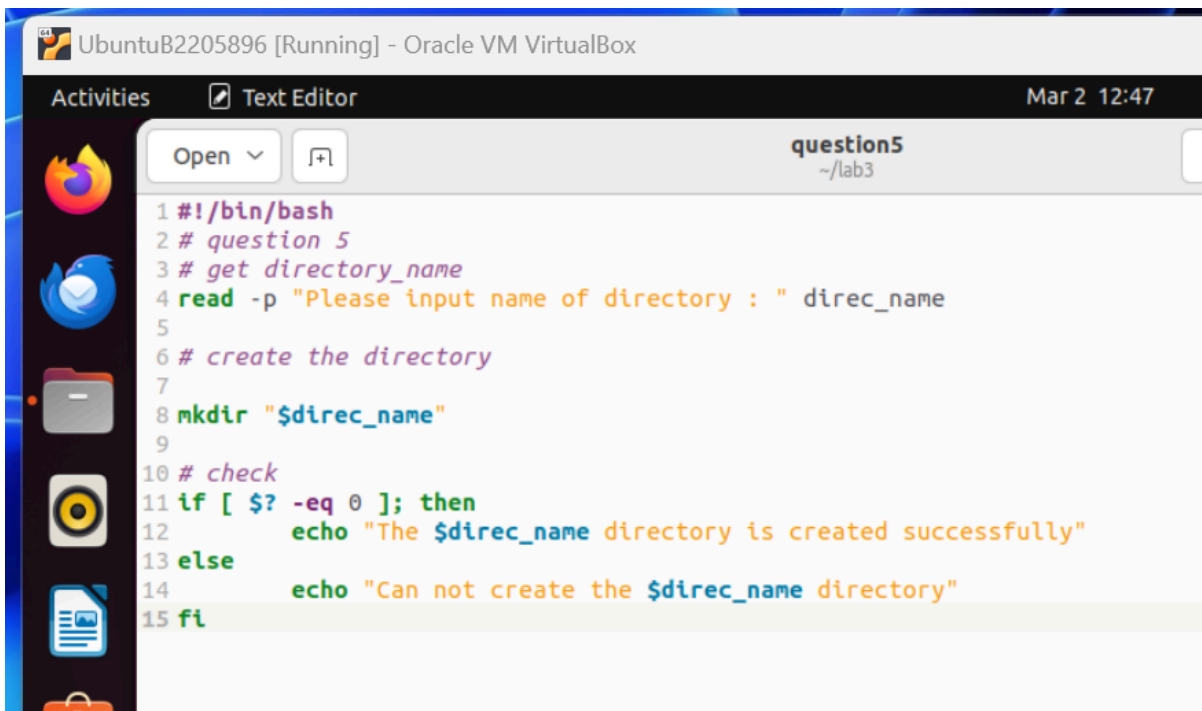
Answer:

```

#!/bin/bash
# question 5
# get directory_name
read -p "Please input name of directory: " direc_name

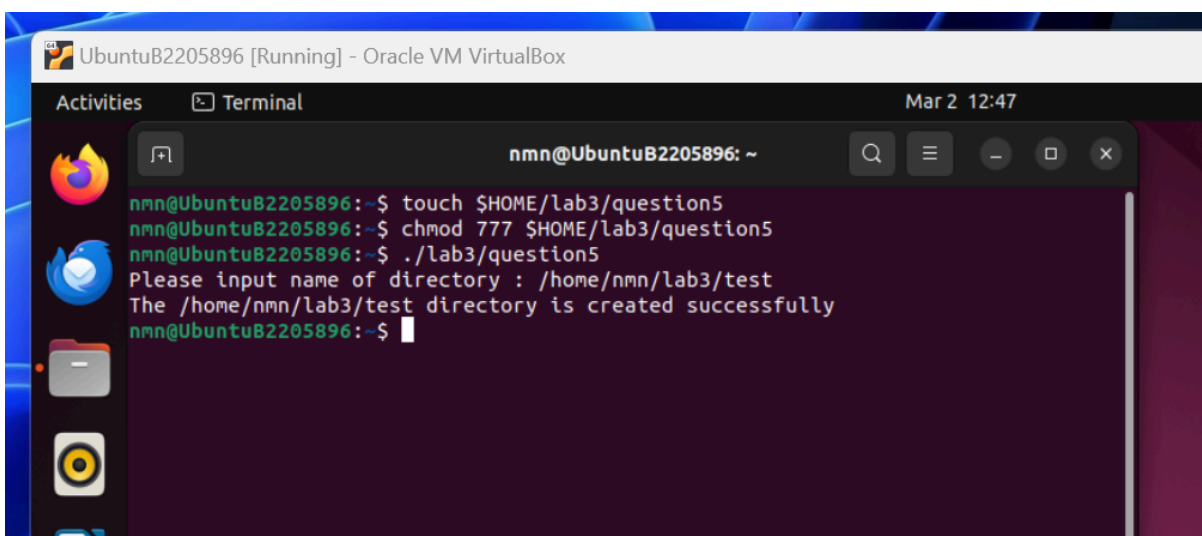
```

```
# create the directory
mkdir $direc_name
# check
if [ $? -eq 0 ]; then
echo "The $direc_name directory is created successfully"
else
echo "Can not create the $direc_name directory"
fi
```



The screenshot shows a text editor window titled 'question5' in the path '~ /lab3'. The script content is as follows:

```
1 #!/bin/bash
2 # question 5
3 # get directory_name
4 read -p "Please input name of directory : " direc_name
5
6 # create the directory
7
8 mkdir "$direc_name"
9
10 # check
11 if [ $? -eq 0 ]; then
12     echo "The $direc_name directory is created successfully"
13 else
14     echo "Can not create the $direc_name directory"
15 fi
```



The screenshot shows a terminal window with the following commands and output:

```
nmn@UbuntuB2205896:~$ touch $HOME/lab3/question5
nmn@UbuntuB2205896:~$ chmod 777 $HOME/lab3/question5
nmn@UbuntuB2205896:~$ ./lab3/question5
Please input name of directory : /home/nmn/lab3/test
The /home/nmn/lab3/test directory is created successfully
nmn@UbuntuB2205896:~$
```

```

nmn@UbuntuB2205896 [Running] - Oracle VM VirtualBox
Activities Terminal Mar 2 12:48
nmn@UbuntuB2205896: ~
nmn@UbuntuB2205896:~$ touch $HOME/lab3/question5
nmn@UbuntuB2205896:~$ chmod 777 $HOME/lab3/question5
nmn@UbuntuB2205896:~$ ./lab3/question5
Please input name of directory : /home/nmn/lab3/test
The /home/nmn/lab3/test directory is created successfully
nmn@UbuntuB2205896:~$ ls $HOME/lab3
question1 question2 question3 question4 question5 test
nmn@UbuntuB2205896:~$
  
```

Question 6: Write a script to print the following symbols to the screen

```

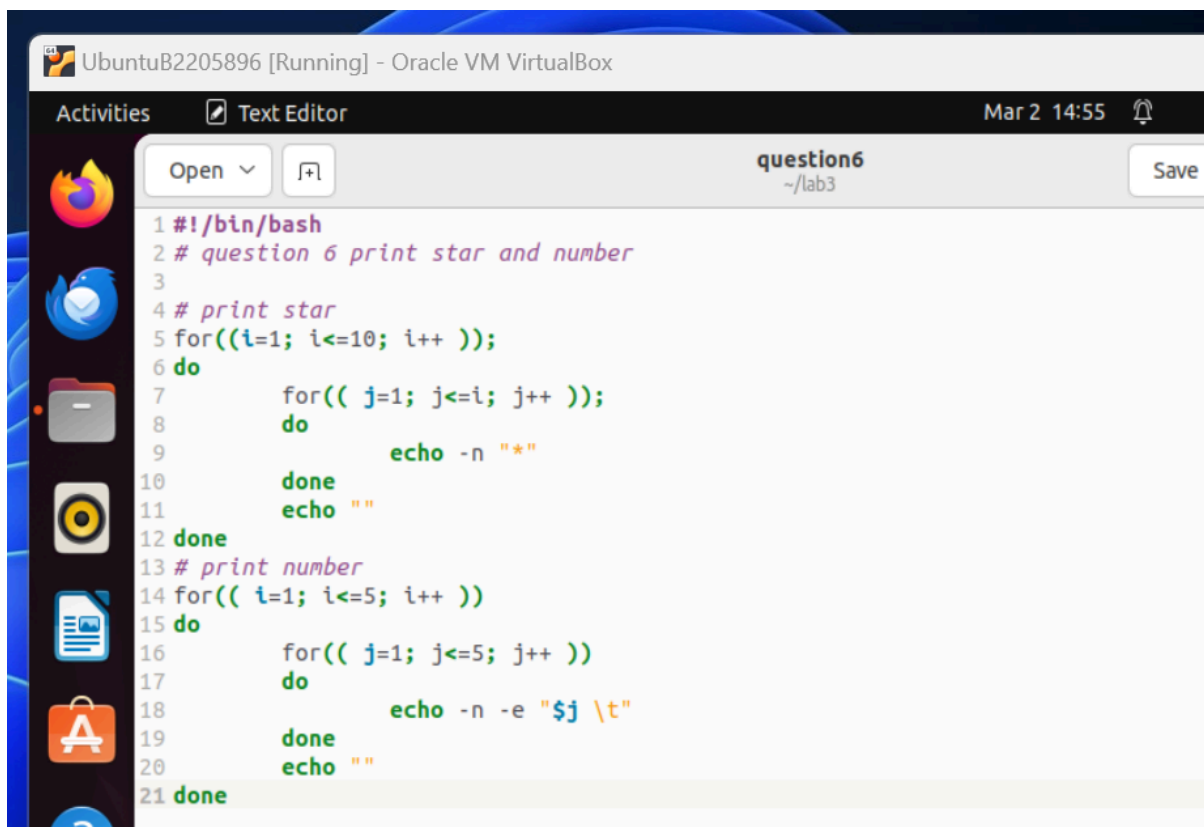
*
**
***
****
*****
*****
*****
*****
*****
*****
*****
1          2          3          4          5
1          2          3          4          5
1          2          3          4          5
1          2          3          4          5
1          2          3          4          5
  
```

Answer:

```

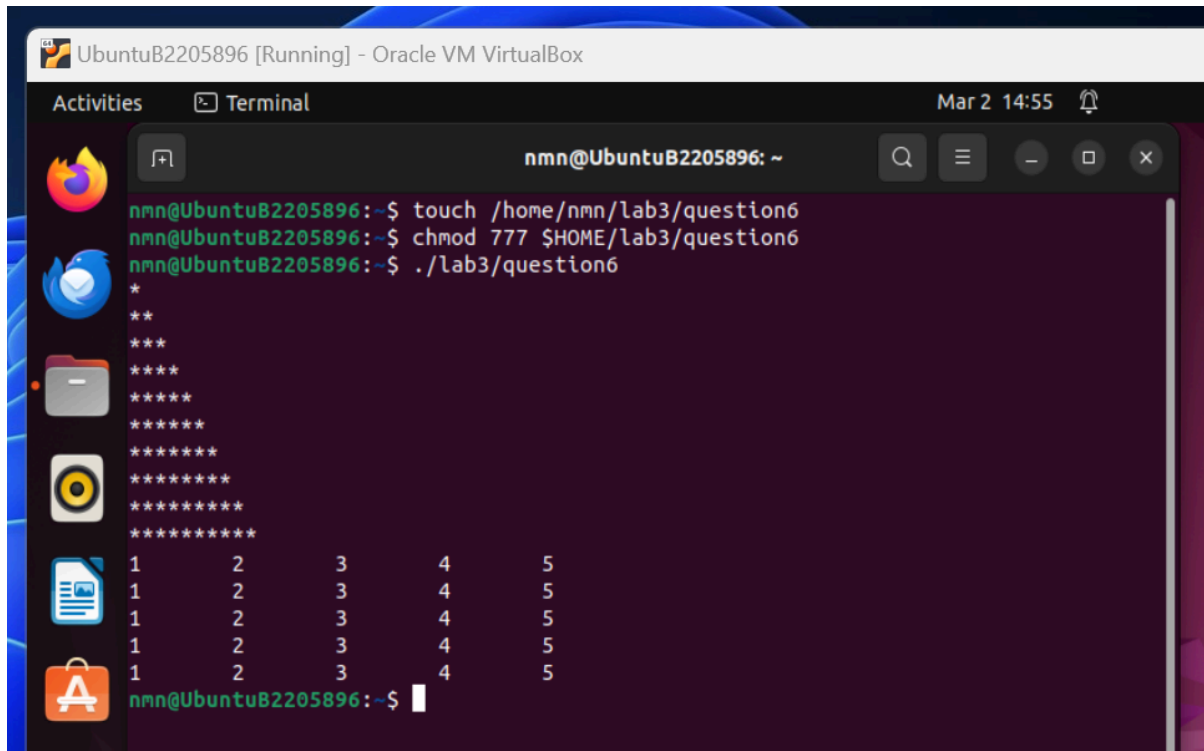
#!/bin/bash
# question 6 print star and number3
# print star
for((i=1; i<= 10; i++ ))
do
for(( j=1; i<=i; j++ ))
do
echo -n "*"
  
```

```
done
echo ""
done
for((i=1; i<=5; i++ ))
do
for(( j=1; i<=5; j++ ))
do
echo -n -e "$j \t"
done
echo ""
done
```



The screenshot shows a text editor window titled "question6" in a virtual machine environment. The window displays a shell script with the following content:

```
1 #!/bin/bash
2 # question 6 print star and number
3
4 # print star
5 for((i=1; i<=10; i++ ));
6 do
7     for(( j=1; j<=i; j++ ));
8     do
9         echo -n "*"
10    done
11    echo " "
12 done
13 # print number
14 for(( i=1; i<=5; i++ ))
15 do
16     for(( j=1; j<=5; j++ ))
17     do
18         echo -n -e "$j \t"
19     done
20    echo " "
21 done
```

```
nmn@UbuntuB2205896:~$ touch /home/nmn/lab3/question6
nmn@UbuntuB2205896:~$ chmod 777 $HOME/lab3/question6
nmn@UbuntuB2205896:~$ ./lab3/question6
*
**
***
****
*****
*****
*****
*****
*****
*****
1      2      3      4      5
1      2      3      4      5
1      2      3      4      5
1      2      3      4      5
1      2      3      4      5
nmn@UbuntuB2205896:~$
```

Question 7: Write a shell script to solve the equation $ax+b=0$ where the coefficients a and b entered from the keyboard

Answer:

```
#!/bin/bash
```

```
# question 7 solve  $ax + b = 0$ 
```

```
read -p "Enter the coefficient a: a"
```

```
read -p "Enter the coefficient b: b"
```

```
if [ $a -eq 0 ]; then
```

```
if [ $b -eq 0 ]; then
```

```
echo "The equation has infinitely many solution"
```

```
else
```

```
echo "The equation has no solution"
```

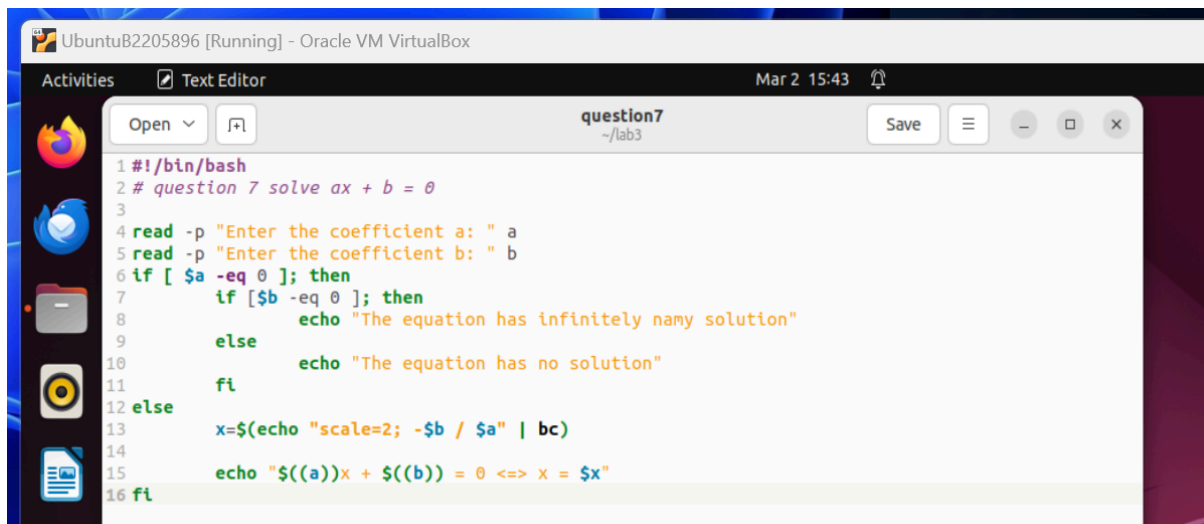
```
fi
```

```
else
```

```
x=$(echo "scale=2; $b / $a" | bc)
```

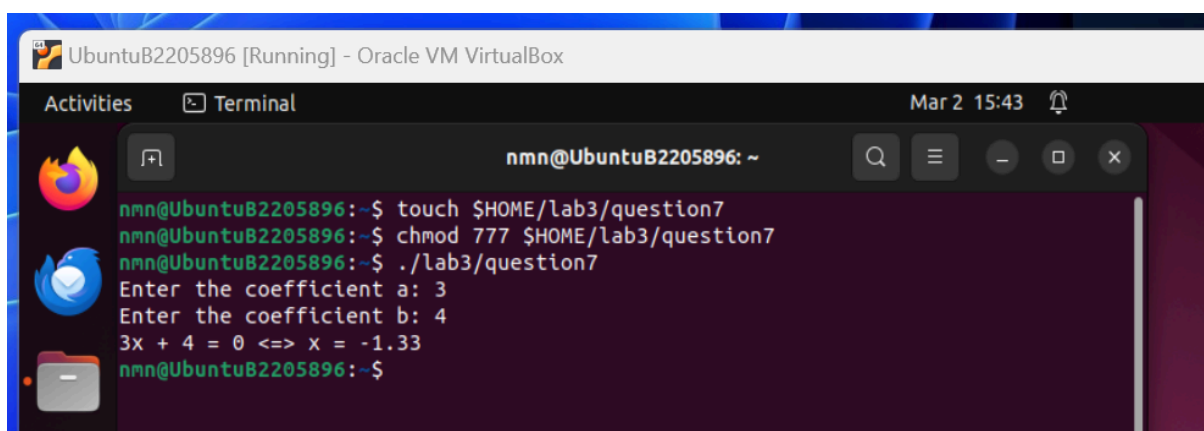
```
echo "$((a))x + $((b)) = 0 <=> x = $x"
```

```
fi
```



```

1 #!/bin/bash
2 # question 7 solve ax + b = 0
3
4 read -p "Enter the coefficient a: " a
5 read -p "Enter the coefficient b: " b
6 if [ $a -eq 0 ]; then
7     if [ $b -eq 0 ]; then
8         echo "The equation has infinitely many solution"
9     else
10        echo "The equation has no solution"
11    fi
12 else
13     x=$(echo "scale=2; -$b / $a" | bc)
14
15     echo "$((a))x + $((b)) = 0 <=> x = $x"
16 fi
  
```



```

nmn@UbuntuB2205896: ~
nmn@UbuntuB2205896:~$ touch $HOME/lab3/question7
nmn@UbuntuB2205896:~$ chmod 777 $HOME/lab3/question7
nmn@UbuntuB2205896:~$ ./lab3/question7
Enter the coefficient a: 3
Enter the coefficient b: 4
3x + 4 = 0 <=> x = -1.33
nmn@UbuntuB2205896:~$
  
```

Question 8: Write a shell script to print a multiplication table of a number entered by the user to the screen. After printing out the multiplication table, the program will ask the user to type “1” if the user wants to continue the process; otherwise, the process of printing the multiplication table will end.

Answer:

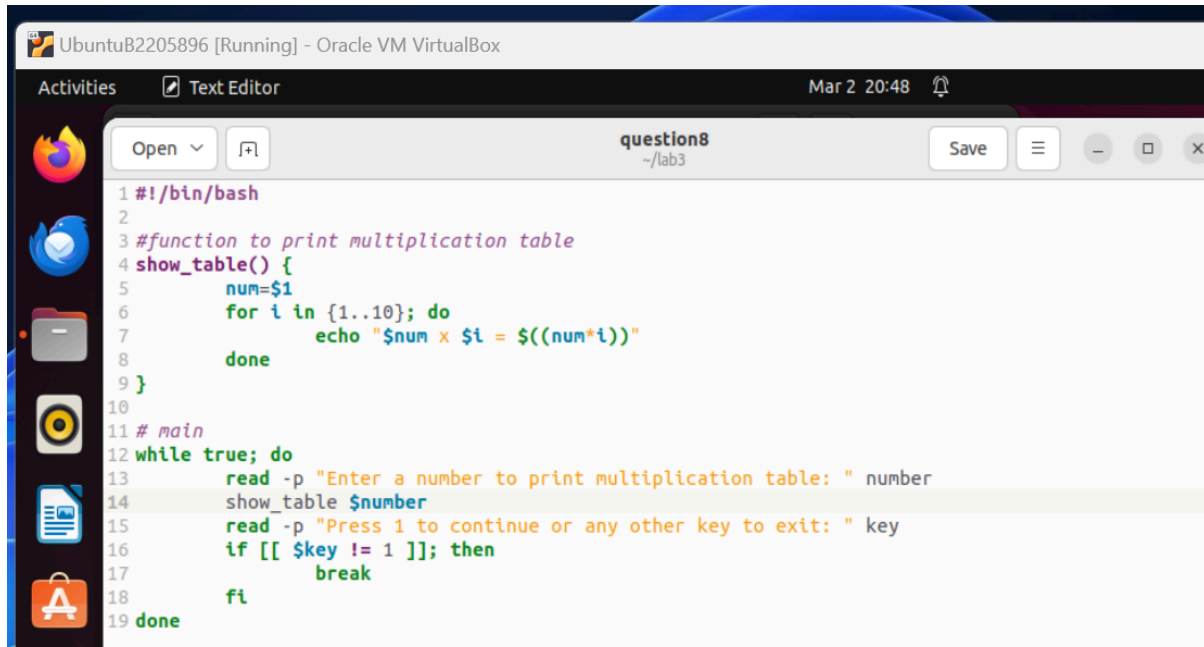
```

#!/bin/bash

#function to print multiplication table
show_table() {
num=$1
for i in [1..10]; do
echo "$num x $i = $((num*i))"
done
}

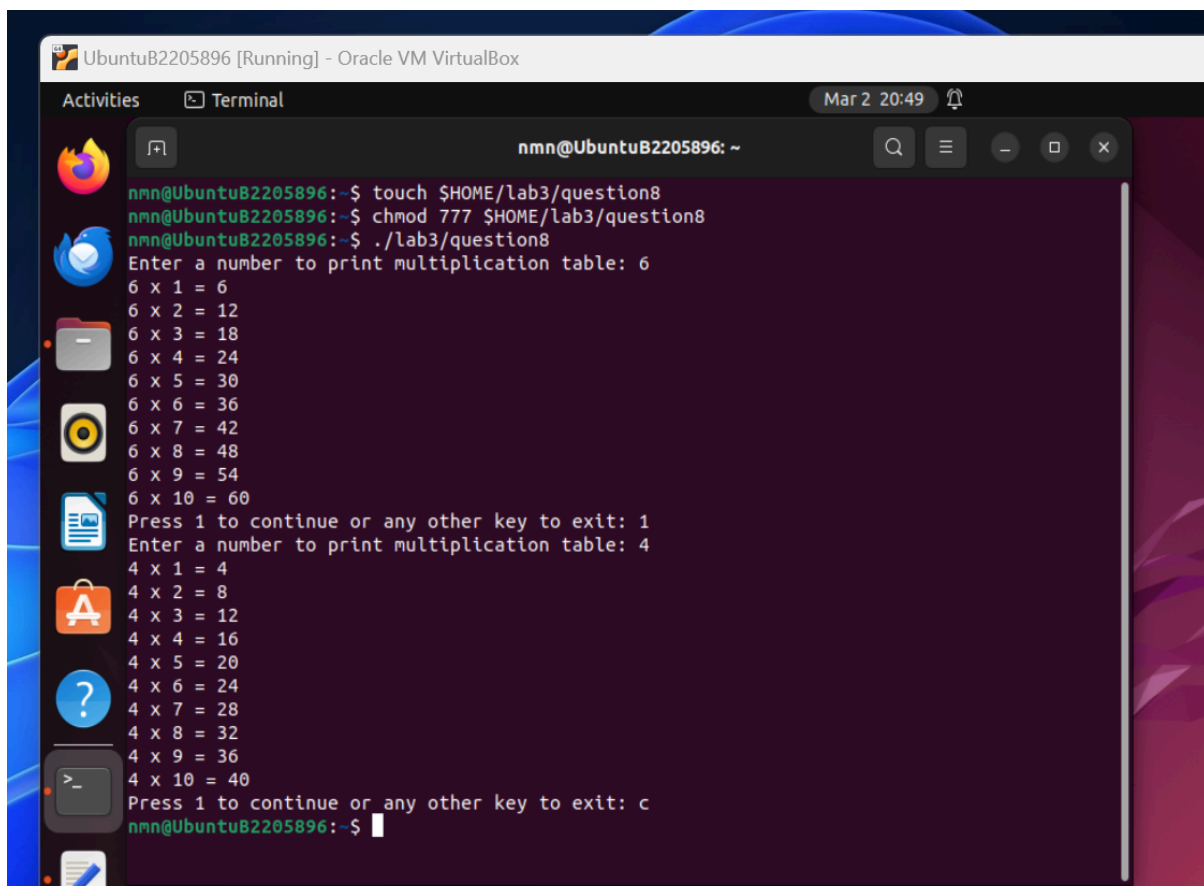
# main
while true; do
read -p "Enter a number to print multiplication table: " number
show_table $number
  
```

```
read -p "Press 1 to continue or any other key to exit: "key
if [[ $key != 1 ]]; then
break
fi
done
```



The screenshot shows a text editor window titled 'question8' in the path '~/lab3'. The script content is as follows:

```
1#!/bin/bash
2
3#function to print multiplication table
4show_table() {
5    num=$1
6    for i in {1..10}; do
7        echo "$num x $i = $((num*i))"
8    done
9}
10
11# main
12while true; do
13    read -p "Enter a number to print multiplication table: " number
14    show_table $number
15    read -p "Press 1 to continue or any other key to exit: " key
16    if [[ $key != 1 ]]; then
17        break
18    fi
19done
```



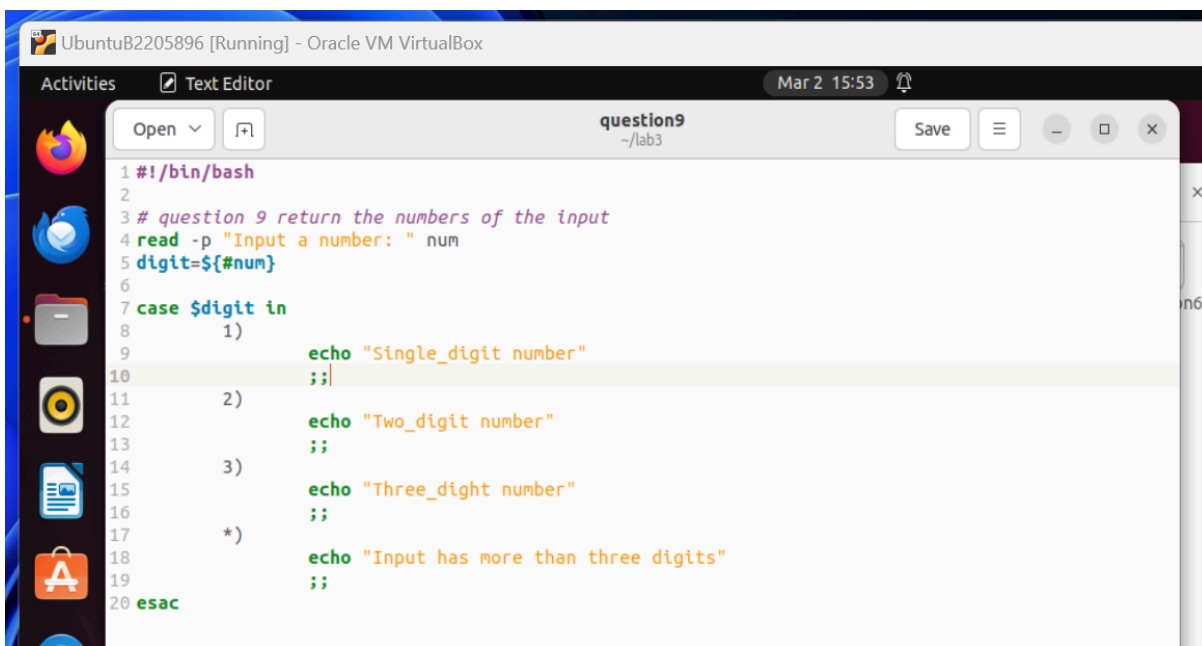
The screenshot shows a terminal window with the following output:

```
nmn@UbuntuB2205896:~$ touch $HOME/lab3/question8
nmn@UbuntuB2205896:~$ chmod 777 $HOME/lab3/question8
nmn@UbuntuB2205896:~$ ./lab3/question8
Enter a number to print multiplication table: 6
6 x 1 = 6
6 x 2 = 12
6 x 3 = 18
6 x 4 = 24
6 x 5 = 30
6 x 6 = 36
6 x 7 = 42
6 x 8 = 48
6 x 9 = 54
6 x 10 = 60
Press 1 to continue or any other key to exit: 1
Enter a number to print multiplication table: 4
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
4 x 6 = 24
4 x 7 = 28
4 x 8 = 32
4 x 9 = 36
4 x 10 = 40
Press 1 to continue or any other key to exit: c
nmn@UbuntuB2205896:~$
```

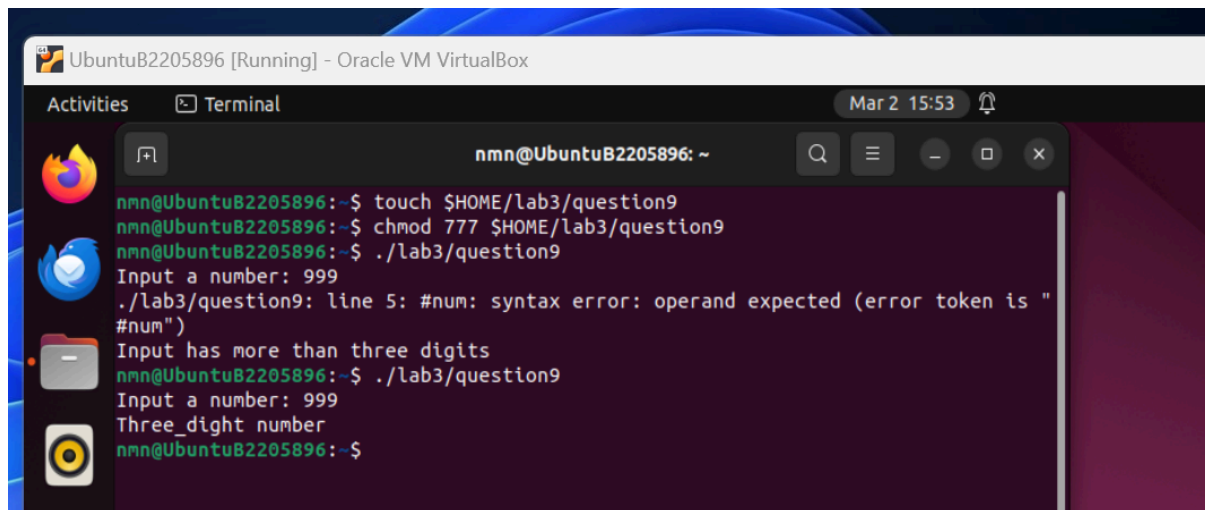
Question 9: Write a shell script that allows users to input an int number and returns the digit number of that number. For example, 10 → returns “two-digit number”; 100 → returns “three-digit number”.

Answer:

```
#!/bin/bash
# question 9 return the numbers of the input
read -p "Input a number: " num
digit=${#num}
case $digit in
1)
echo "Single_digit number"
;;
2)
echo "Two_digit number"
;;
3)
echo "Three_dight number"
;;
*)
echo "Input has more than three digits"
;;
esac
```

A screenshot of a text editor window titled "question9" in a virtual machine environment. The window shows the shell script code from the previous block. The code is color-coded: comments are green, keywords like 'read', 'case', and 'esac' are blue, and strings are orange. The window has a standard Ubuntu interface with a sidebar showing application icons and a top bar with system information like "Mar 2 15:53".

```
1 #!/bin/bash
2
3 # question 9 return the numbers of the input
4 read -p "Input a number: " num
5 digit=${#num}
6
7 case $digit in
8     1)
9         echo "Single_digit number"
10        ;;
11    2)
12        echo "Two_digit number"
13        ;;
14    3)
15        echo "Three_dight number"
16        ;;
17    *)
18        echo "Input has more than three digits"
19        ;;
20 esac
```



```
nmn@UbuntuB2205896:~$ touch $HOME/lab3/question9
nmn@UbuntuB2205896:~$ chmod 777 $HOME/lab3/question9
nmn@UbuntuB2205896:~$ ./lab3/question9
Input a number: 999
./lab3/question9: line 5: #num: syntax error: operand expected (error token is "
#num")
Input has more than three digits
nmn@UbuntuB2205896:~$ ./lab3/question9
Input a number: 999
Three_dight number
nmn@UbuntuB2205896:~$
```

-----END-----